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Carbon Accounting and Reporting

Lecture 3 – Introduction to Organisational Carbon Accounting

Dr James Robey / Jon Hampson
26th September 2025

Carbon Accounting Exercise

Step 1 – UK Government “DEFRA” emissions factors

1. For this exercise, you will need to download the DEFRA emission factor
<https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2025>
2. Choose Greenhouse gas reporting: conversion factors 2024 and then download “Conversion factors 2025: condensed set (for most users)”
3. Spend a few minutes to familiarize yourself with the spreadsheet

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Research and analysis

Greenhouse gas reporting: conversion factors 2025

These emission conversion factors are for use by UK and international organisations to report on 2025 greenhouse gas emissions.

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Documents



[Conversion factors 2025: condensed set \(for most users\)](#)

MS Excel Spreadsheet, 1.75 MB

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[Request an accessible format.](#)

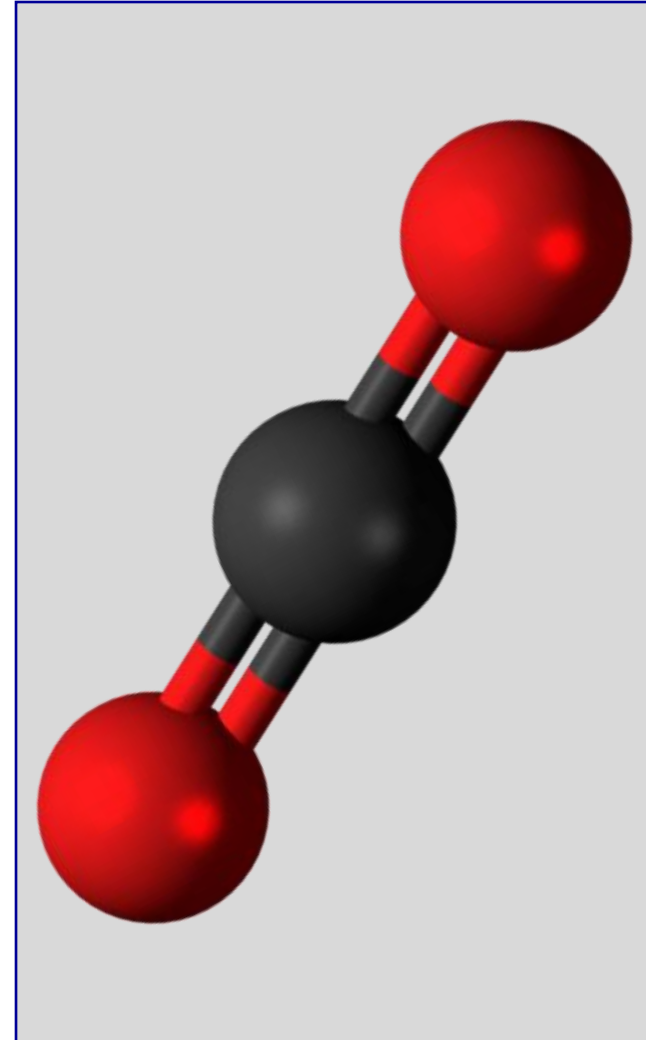
Related content

Collection

[Government conversion factors for company reporting of greenhouse gas emissions](#)

Step 2 – Shark Consulting Limited

- Shark Consulting Limited is a small consulting company with 15 employees that provide consulting services to clients around the UK and Europe
- They operate from a single office in London and from various client sites around the UK and Europe
- The owners of the business, Sarah and Simon, have decided to calculate the carbon footprint of their business and have collected a range of data points
- Based upon the data over the next few pages and what you have learnt so far, use the 2025 DEFRA emission factors, please calculate their carbon footprint for 2024



Shark Consulting – data points – energy

- Shark Consulting has one office.
- The following data is from the monthly energy bills for electricity and natural gas

Monthly electricity bill				Monthly gas bill (natural gas)			
	kwh	GBP			cubic metres	GBP	
Jan-24	2,450	£	735.00	Jan-24	460	£	322.00
Feb-24	2,750	£	825.00	Feb-24	420	£	294.00
Mar-24	2,350	£	705.00	Mar-24	375	£	262.50
Apr-24	2,500	£	750.00	Apr-24	250	£	175.00
May-24	2,100	£	630.00	May-24	125	£	87.50
Jun-24	1,750	£	525.00	Jun-24	110	£	77.00
Jul-24	1,800	£	540.00	Jul-24	100	£	70.00
Aug-24	1,200	£	360.00	Aug-24	75	£	52.50
Sep-24	1,500	£	450.00	Sep-24	150	£	105.00
Oct-24	1,750	£	525.00	Oct-24	225	£	157.50
Nov-24	2,250	£	675.00	Nov-24	385	£	269.50
Dec-24	1,950	£	585.00	Dec-24	415	£	290.50
	24,350	£	7,305.00		3,090	£	2,163.00

Shark Consulting – data points – car travel

- Sarah and Simon each have a 'luxury' car which belongs to Shark Consulting Limited
- Six other employees drive 'executive' cars provided through a third-party leasing company (2 are petrol, 2 are diesel, and 2 are plug-in hybrids)
- The other employees exclusively use public transport

			Excutive Cars (2 Petrol, 2 Diesel, 2 Hybrid)			
2 Luxury Cars (Petrol)				combined km driven		
combined km driven				Petrol cars	Diesel cars	Hybrid cars
Jan-24	2,515		Jan-24	1,856	2,566	1,622
Feb-24	2,355		Feb-24	2,566	2,368	2,655
Mar-24	3,215		Mar-24	2,400	2,566	3,258
Apr-24	4,518		Apr-24	2,356	2,955	1,622
May-24	2,655		May-24	2,900	2,145	2,955
Jun-24	1,544		Jun-24	1,595	2,714	2,864
Jul-24	1,860		Jul-24	1,653	1,862	3,651
Aug-24	252		Aug-24	896	1,153	1,205
Sep-24	2,654		Sep-24	1,799	1,836	1,985
Oct-24	3,502		Oct-24	1,865	1,955	2,841
Nov-24	2,806		Nov-24	1,732	1,266	2,315
Dec-24	1,524		Dec-24	1,862	1,725	1,985
Total	29,400		Total	23,480	25,111	28,958

Shark Consulting – data points – air travel

- Shark Consulting Limited employees took the following flights during 2024
- All flights were in Europe except for one trip to New York undertaken by Simon and Sarah to attend a conference
- All fights were in economy except for Sarah's return leg from New York where she used air miles to upgrade to business class

Air Travel		
	Distance km	Return trips
London (LHR) / Paris (CDG)	348	6
London (LHR) / Rome (ROM)	1,446	15
London (LHR) / Amsterdam (AMS)	372	16
London (LHR) / Krakow (KRK)	1,432	3
London (LHR) / Madrid (MAD)	1,243	9
London (LHR) / Gothenburg (GOT)	1,071	14
London (LHR) / Zurich (ZRH)	790	16
London (LHR) / Helsinki (HEL)	1,853	8
London (LHR) / New York (JFK)	5,555	2

Shark Consulting – data points – other travel

- Shark Consulting employees have estimated that they travelled the following distance on trips during 2024

Other travel	
	Distance km
Taxis (Black cab)	4,200
Taxis (other)	7,400
Bus (London)	7,850
National Rail	132,000
London Underground	17,600

Questions (Step 1 of 2)

Help Sarah and Simon to calculate their 2024 carbon footprint

1. What is Shark Consulting's carbon footprint for 2024 based on the data provided?
2. Which is larger – the scope 1, 2 or 3 emissions?
3. How could Shark Consulting improve the calculation of their carbon footprint?

E.g. Are there any other emissions sources they should investigate?
Are there any opportunities to improve their data?

(Write down any assumptions that you have made.)

Questions (Step 2 of 2)

Sarah and Simon are looking to implement a decarbonisation programme. They are interested to explore the impact of a number of initiatives.

Calculate what the impact on their 2024 would have been if they had implemented the following carbon saving ideas.

4. Replace all their cars (owned and leased) with Battery Electric Vehicles
5. Used the Eurostar for all the trips to Paris and Amsterdam (rather than flying)

(Write down assumptions that you have made.)

Please email your solutions (one per group) by 16:00 on Thursday 2nd October
(remember to include your group number)

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